MIL-E-1/1675(EL)
18 July 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS-SWITCHING
TYPE 6795A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, X-Band (Band Pass), 9000 to 9400 MHz.

OUTLINE AND DIMENSIONS: See Fig. 1 herein. WEIGHT: 2.2 oz (approx).

APPLICATION: Branched duplexer; terminated with Semiconductor Diode 1N23WG mounted in appropriate crystal holder; for Waveguide RG-52/U mounting.

MOUNTING: Either end of this electron tube may be used for input or output connection respectively.

RATINGS: (Note 1)

Parameter:	Transm. Po	Duty Cycle	Freq.	Open Ckt. Ign. Volt	Ign. Curr.	Oper. Temp.	Oper. Alt.
Unit:	kw	Du	MHz	Vdc	uAdc	°C	ft.
Maximum	40	.002	9400	---	200	+100	35,000
Minimum	4.0	---	9000	-650	100 (Note 2)	-40	---

Method	Requirement or Test	Conditions	AQL (% defec.)	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
	GENERAL							
---	Preproduction Sample Approval	Required;(Note 3)	---	---	---	---	---	---
---	Marking	(Note 4)	---	---	---	---	---	---
---	Dimensions	(See Fig. 1)	---	---	---	---	---	---
---	Packaging	(Notes 5,6)	---	---	---	---	---	---

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Method	Requirement or Test	Conditions	AQL (% defec.)	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
	QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 7)							
---	Visual-mechanical examination	See Fig. 1	---	---	---	---	---	---
4473	Voltage standing-wave ratio	$\sigma' = 1.05$ max (Note 8) F1 = 9,000±0.1%MHz F2 = 9,200±0.1%MHz F3 = 9,400±0.1%MHz	0.65	II	σ	---	1.4	(ratio)
4451	Flat leakage power	F = 9,200±0.5%MHz Po = 40 KW±10% $I_{(ign)}$=100 uAdc tp1=1.0±0.15usec tp2=0.5±0.15 usec prr=1000 pps σ=1.10 max (Notes 9,10)	0.65	II	pf	---	40	mW
4446	Spike leakage energy	(Same conditions as for Flat Leakage Power test above)	0.65	II	Ws	---	0.2	erg
4401	Ignitor ignition time	Ebb= -700 Vdc R1 = 5.5 megohms	0.65	II	t	---	5	sec.
4488	Arc loss	po = 4.0 KW tp = 1.0 usec prr = 1000 pps	0.65	II	La	---	0.8	db
4406	Ignitor voltage drop	$I_{(ign)}$ = 100 uAdc	0.65	II	Eid	-200	-375	Vdc
4471	Recovery time	Method A F = 9,200±0.5%MHz Po = 40 KW±10% tp = 1.0 usec prr =1000 pps $I_{(ign)}$=100 uAdc (Notes 9,11)	0.65	II	t	---	10	usec

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Method	Requirement or Test	Conditions	AQL (% defec.)	Insp Level or Code	Symbol	Limits Min	Limits Max	Unit
	QUALITY CONFORMANCE INSPECTION, PART 2							
4421	Ignitor interaction	$I_{(ign)}$=100 uAdc	6.5	S3	ΔLi	---	0.2	db
---	Ignitor noise ratio	(Note 12) F = 9,200±0.5%MHz $I_{(ign)}$= 100 uAdc	6.5	S3	Nr	---	1.3	(ratio)
4416	Insertion loss	F=9,000 to 9,400 MHz, band swept $I_{(ign.)}$ = 0 uAdc	6.5	S3	Li	---	0.7	db
	QUALITY CONFORMANCE INSPECTION, PART 3 LIFE							
---	Life test	Group D F=9,200±0.5%MHz Po =40 KW±10% tp =1.0±0.15 usec prr = 1000 pps $I_{(ign)}$ = 100 uAdc (Notes 9,13,14)	---	--	t	500	---	hrs
	Life Test end-points:	F = 9,200±0.5% MHz						
		NF degradation of test diode			NF	---	+3	db
		Recovery time			t	---	20	usec
		Insertion loss			Li	---	0.7	db
		Flat leakage power			pf	---	50	mW
		Spike leakage energy			Ws	---	0.25	erg

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Method	Requirement or Test	Conditions	AQL (% defec.)	Insp Level or Code	Symbol	Limits Min	Limits Max	Unit
	QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC-CHECK, (NOTES 15, 16)							
1002	Barometric pressure (reduced)	Press.=178 mm Hg $R_{(ign)}$ =5.5 megohms Ebb = -700 Vdc	---	---	---	---	---	---
4021	Vibration	10G	---	---	---	---	---	---
1027	Temperature cycling	T = -40°C to +100°C	---	---	---	---	---	---
1011	Humidity	---	---	---	---	---	---	---
1042	Shock	Test Cond. A	---	---	---	---	---	---

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NOTES:

1. Refer to Specification MIL-E-1, Section 6 "Note" covering absolute ratings.

2. Recommended ignitor operating current = 150 uAdc; this is for a tube having an average ignitor voltage drop. The following formula should be used to determine the value of the series resistance during test or equipment-circuit application of the tube:

$$R = \frac{Ebb - Eid}{150}$$

where: R = total series resistance (megohms)
Ebb = open-circuit supply voltage (volts)
Eid = average (center) ignitor voltage drop (volts)

At least 0.5 megohms of the total resistance should be located as close as possible to the ignitor top cap, to prevent oscillation.

3. Preproduction Sample Approval in lieu of Qualification.- Approval of preproduction sample, based upon preproduction sample inspection, shall replace Qualification requirements of Specification . . . . MIL-E-1. Preproduction Sample tests, testing, sampling, and approval criteria shall correspond directly with all pertinent requirements in the specification as normally applicable to Qualification of product.

   a. Preproduction tests will be performed after award of contract, at the manufacturer's plant, unless otherwise specified in the bid request or contract. Approval of the preproduction sample is required prior to initial delivery of units on the contract. The fabrication of production units, prior to approval of the preproduction sample, shall be at the manufacturer's risk. Unless otherwise specified in the bid request or contract, the specimens shall become the property of the Government and, upon approval, shall serve as a reference standard for the contracting officer and Government inspector.

   b. Requests for information regarding preproduction testing and approval of the product(s) covered herein should be referred to the Commanding General, U. S. Army Electronics Command, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, attention: AMSEL-PP-EM-1.

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c. No Qualified Products List (QPL) entry will be undertaken for the product(s) covered herein as approval of design and performance of product will be based upon Preproduction Sample Approval.

d. In no manner, shall the above requirements for Preproduction Sample Approval be construed as affecting requirements for production acceptance inspection of the product(s) covered herein.

4. Tube-designation marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only. (See Fig. 1 herein.)

5. Each electron tube shall be furnished with dust covers for protection of terminal-connector elements.

6. Unless otherwise specified in contract or order, each tube shall, as a minimum protection against moisture and moisture vapor, be sealed within a moisture-vaporproof barrier bag, or equivalent, normally acceptable under packaging requirements of Specification MIL-E-75.

7. The AQL for the combined defectives for attributes in Quality Conformance Inspection, Part 1, exclusive of "Visual-Mechanical examination", shall be 1.0%. A tube having more than one defect shall be counted as one defective.

8. Although either end of tube may be used as the input end, test at any one of the specified frequencies shall be performed at "input" end opposite to that used for the other required frequencies.

9. This test shall be performed with the electron tube terminating the applicable transmission line.

10. The Flat Leakage Power and Spike Leakage Energy measured shall be within the specified limits when they are determined by the following (associable) procedure(s):

a. The Leakage Power is monitored through a directional coupler and video crystal-detector, and displayed on an oscilloscope screen. The input impedance of the oscilloscope shall match the crystal-detector impedance. The Leakage-Power pulse shape may be of a form as shown in Figure 2 herein, where t1 and t2 are the Leakage-Power pulse widths corresponding to the specified transmitter pulse width of tp1 and tp2.

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NOTES-(Cont'd).

b. A reference is established on the oscilloscope screen by previously calibrating the crystal detector and oscilloscope with a known pulse or CW power with a peak value corresponding to the maximum values of Flat Leakage Power specified. If the peak of the flat portion of the Leakage Pulse corresponding to t2 is less than the calibrated reference level, the Flat Leakage Power requirements are satisfied.

c. The Spike Leakage Energy is determined from the value P1 (the total leakage power at pulse width, tp1) if the following equation-condition exists:

$$P1 < \frac{(Ws)\ \ Max.\ \ (prr)}{10^7} \qquad (1)$$

d. Where (Ws) Max. (see c, above) is the maximum value of Spike Leakage Energy specified, the Spike Leakage requirements are satisfied. If equation (1) is not satisfied, then the Spike Leakage Energy is determined from:

$$Ws = \frac{10^7}{prr}(P1 - Wf1) \qquad (2)$$

where Wf1 is the energy of the flat power contained in t1, and is determined by analyzing the leakage power pulse shape.

e. For simplicity, it may be desirable to reduce the specified value tp1 such that the energy in the Flat Power Wf1 is negligible. In this case, Equation (1) is directly applicable for computing Ws, and the value of tp1 in the standard must be changed accordingly.

11. Recovery time shall be measured at end of 30 minutes of operation of the tube under the specified conditions.

12. Using a circuit per Fig. 3 herein, the I.F. attenuation required to reduce the output noise level with the ignitor ON, to that with the ignitor OFF, shall be determined. The ignitor noise ratio , (Nr), is calculated from the following equation:

$$Nr = 1 + \frac{(\text{antilog } 0.1F) \times [(\text{Antilog } 0.1Y) - 1]}{2}$$

where: F = Receiver noise figure in db
Y = I.F. attenuation required to reduce the output noise level with the ignitor ON to that with the ignitor OFF.

The factor 1/2 in the second term is applicable only if no preselector is used. Curves for noise ratio as a function of I.F. attenuation are shown in Figure 4 herein.

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13. The electron tube under test shall be terminated with Semiconductor Diode type 1N23WG mounted in a suitable waveguide crystal holder. Diode 1N23WG shall conform to the requirements of the latest issue of Specification MIL-S-19500/322.

14. The following additional Life-test requirements and criteria shall apply:

    a. A 100-ohm video load mount shall be employed, placed directly behind the tube.

    b. At beginning of Life test, each crystal diode shall meet the minimum noise-figure requirements specified in its applicable specification (see Note 13 above). Throughout the duration of Life test, the crystal-diode noise Figure shall be periodically remeasured, and:

        (1) The Life test longevity shall be considered completed either at the end of the specified time, or when the degradation in the crystal-diode noise figure has exceeded the allowable amount.

        (2) One crystal-diode failure on only one tube of the entire Life-test sample will be allowed. Should such a failure occur before the end of the Life test, a new crystal diode may be inserted, and the Life test continued on that one tube. Hereto, the (renewed) Life test shall not be terminated before satisfactory protection by the particular tube, for a minimum of 100 hours, is demonstrated on the replacement crystal diode, even if the Life test must be extended beyond the specified time. For example: if, during the 500-hour Life test crystal-diode failure occurs at 477 hours, a new crystal diode would be inserted and the test would continue for an additional 100 hours even though the 500-hours time would be exceeded by 77 hours.

15. For periodic tests, sample tubes are to be selected at random intervals so that during any twelve-month period a minimum of two tubes shall have been successfully tested (see Note 16 below). In the event of failure of one or more tubes in a sample, these tests are to become a part of Quality Conformance Inspection, part 2, until two successive lots have been successfully completed on these tests. After two successive lots have thus been successfully completed, the periodic tests shall then be eligible to be resumed on a twelve-month basis.

16. After subjection to all Periodic-check tests, the sample tube(s) shall meet the requirements of all tests in QCI, Part 1 herein.

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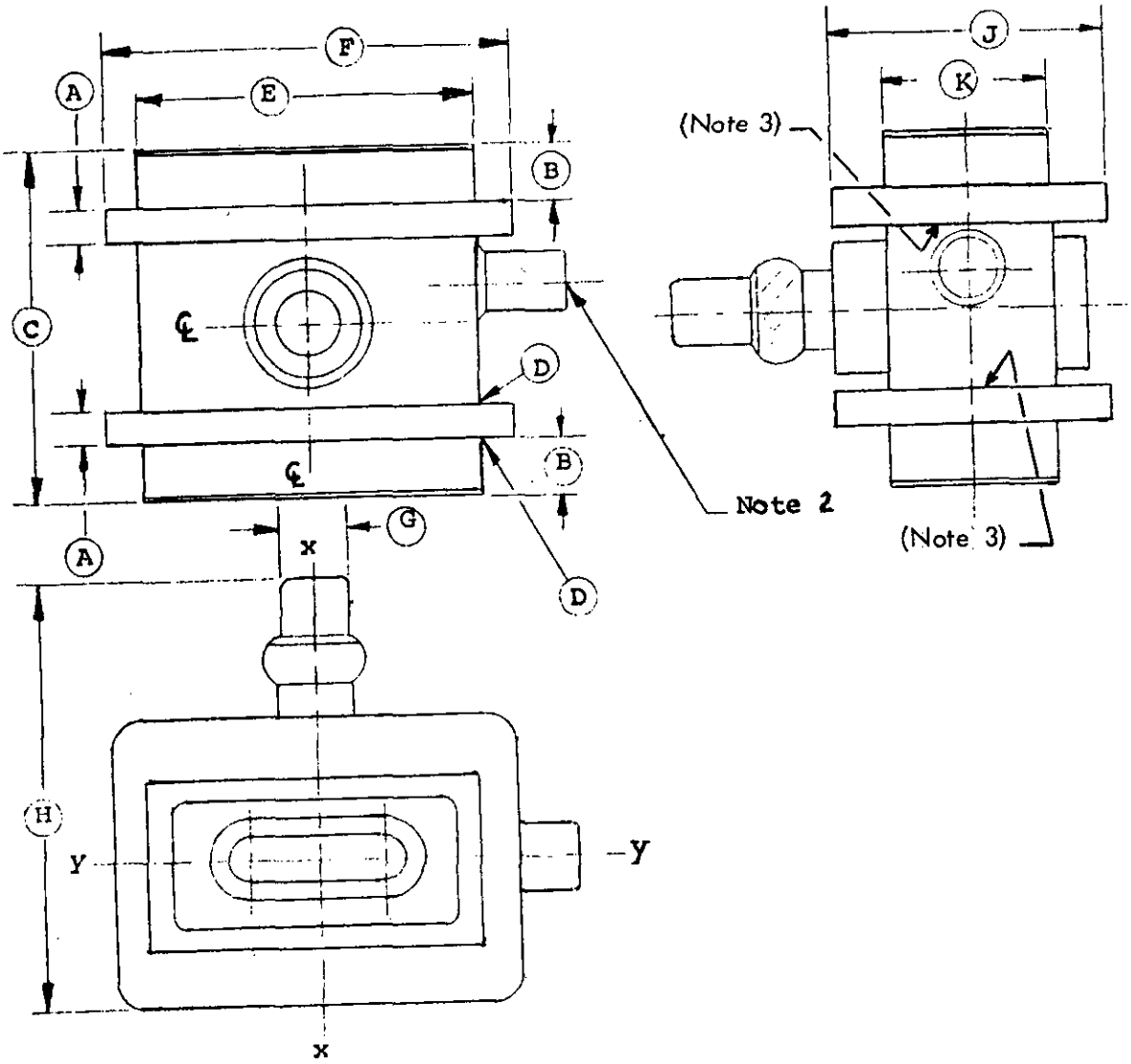


Figure 1. Outline and dimensions.

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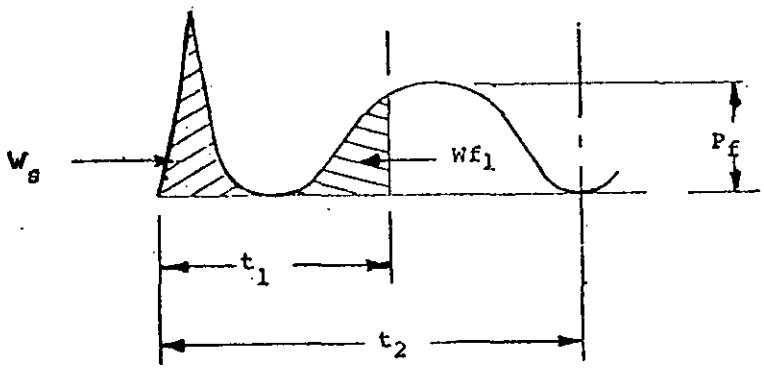
Ltr	Minimum	Maximum
PREPRODUCTION SAMPLE APPROVAL		
A	.092(2.336)	.102(2.590)
E	.995(25.273)	1.010(25.654)
F	1.195(30.353)	1.205(30.607)
G	.245(6.223)	.255(6.477)
J	.695(17.653)	.705(17.907)
K	.495(12.573)	.510(12.954)
QUALITY CONFORMANCE INSP. (PART 1)		
B	.180(4.572)	.190(4.826)
C	1.011(25.679)	1.021(25.933)
D	---	.030(0.762)
QUALITY CONFORMANCE INSP. (PART 2)		
H	---	1.312(33.324)

NOTES:

1. Dimensions in inches with metric equivalents (mm) in parentheses.
2. Exhaust stem not to extend more than 0.187 in. beyond flanges.
3. Temperature measurement point: on flange surface at junction of tube body, at "input" end of tube.
4. The AQL for combined mechanical defectives in QCI-1 shall be 1.0%.
5. The AQL for combined mechanical defectives in QCI-2 shall be 6.5%, Insp. Level S-3.
6. Finishes: Plating = cadmium (0.0003 in., min.) over copper plate; No paint.
7. The tube identification and serial number shall be marked in a location on each tube most practical for ready discernment when mounted for either-end "input" application.

Figure 1 - (Cont'd). Outline and dimensions.

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W_s = Value of spike leakage

Wf_1 = Energy of flat power contained in t_1.

P_f = Flat Leakage Power

t_2 = Leakage Power pulse width corresponding to the specified transmitter pulse width of tp_2

t_1 = Leakage Power Pulse width corresponding to the specified transmitter pulse width of tp_1

Figure 2. Pulse-Shape relative to Leakage Power Test.

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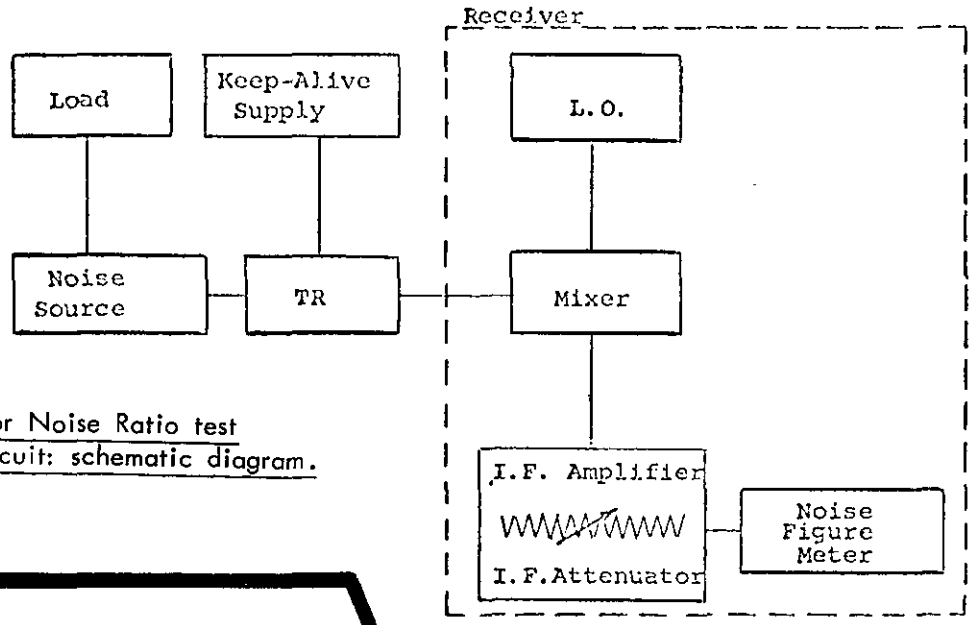


Fig. 3. Ignitor Noise Ratio test circuit: schematic diagram.

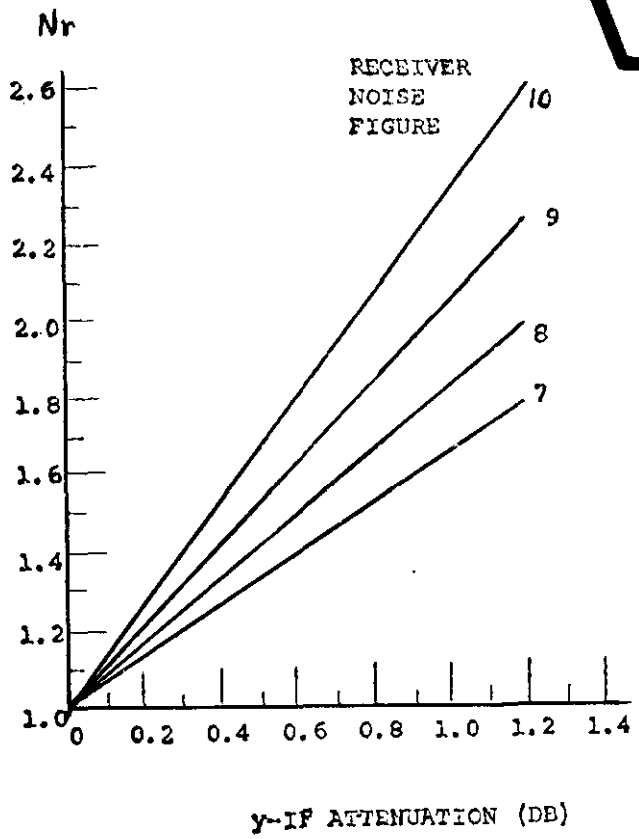


Fig. 4. Noise Ratio vs. IF attenuation: graph.

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